

الحسابات الخاصة باللولب

الملح = ناعم
كثافة الملح = ١.٣ جرام / سم^٣

أبعاد اللولب = [الطول cm] × [العرض cm]

الوزن المحدد حالياً هو 500 جرام
بعد الإنتاج المستهدف [15] خمسة عشر كغ
15 كغ / دقيقة

قوة الورد = 20 Kg

نسبة الفشل: [ورشة صقلية 7 - 25 °C]
الزلاوية 40 %

$$384.6 = \frac{500}{1.3} = \frac{m}{\rho} = \frac{500}{1.3}$$

$$\text{مساحة وجه اللولب} = 12 \times 20 = 240 \text{ cm}^2$$

$$\text{مك اللولب} = \frac{\text{الحجم}}{\text{المساحة}} = \frac{384.615 \text{ cm}^3}{240 \text{ cm}^2} = 1.60$$

$$\text{مساحة مقطع اللولب} \\ A = \frac{\pi D^2}{4} = \frac{\pi (0.06)^2}{4} = 2.827 \text{ mm}^2$$

مساحة المقطع

سرعة الالول لكل الالول

$$V_{rev} \approx A \cdot P \cdot FF$$

↓ Pitch (خط الالول)
Fill factor

$$A = 2.827 \text{ mm}$$

$$P \approx \text{Pitch} = 60 \text{ mm}$$

$$FF = \frac{\text{الكثافة الظاهرة}}{\text{الكثافة الحقيقية}} = \frac{1.3}{2.17} = 60\%$$

$$\Rightarrow V_{rev} = A \cdot P \cdot FF$$

$$= 0.2827 \text{ mm} \times 60 \text{ mm} \times 6$$

$$V_{rev} \approx 101.772 \text{ cm}^3$$

الدورات المظلمة = الحجم المظلم

تخم الالول لكل دورة

$$3.8 \text{ دورة} = \frac{384.62}{101.78}$$

سرعة الدوران = 60 revs-per-fill

Fill time (s)

Helix angle at mean diameter

$$\phi = \arctan\left(\frac{P}{\pi D_m}\right) = 26.6^\circ$$

$$\therefore D_m = \frac{D + D_{\text{shaft}}}{2} = \frac{60 + 16}{2} = 38 \text{ mm}$$

Servo motor 200W - 300, 350W -

Pitch = 60 mm

fill factor $\cong 60\%$

flight plate thickness $\cong 4$ mm

clearance (tip-to-housing) $\cong 1.5$ mm

عدد اللفات الواقعة داخل
الولب عند الحلزون تقريبا
أربع لفات عن طول flight

Radial shaft height of flight $\cong 20$ mm

flight overhang tolerance at
each end leave clearance $\cong 10$ mm

Helix angle زاوية الحلزون
 $\cong 26.69^\circ$

Flight length (effective) ≈ 228 mm
Total " " $\approx 300-350$ mm

Number of turns ≈ 3.8 rev

Clearance tip to rough $\approx 1.5 \text{ mm}$

$RPM \stackrel{N}{=} 113.37 \text{ rpm}$

$$W = 11,868$$

$$T = 6 - 15 \text{ N}\cdot\text{m}$$

$$sf \Rightarrow$$

$$P_{\text{motor}} = SF \times P_{\text{required}} \quad \underline{N} \geq 118.69$$

$$\underline{N} = 356.07 \text{ W}$$

$$P_{req} = T_{working} \times W \approx 10 \times 11.868$$

$$\underline{\underline{\cong 118.68 \text{ W}}}$$

Servo motor 200 - 350 W

flight diameter ≈ 90 mm

Shaft $\approx 20 \text{ mm}$

Overall $\approx 60 - 65 \text{ mm}$

طول الخط الذي يتحركه الملح 228 mm

السرعة الدورانية 113.37 RPM

عزق التقديرى $N.m 15 - 6 = T$
 $N.m 6 \approx \text{Working}$

flight outer diameter = 40 mm

shaft " " = 20 mm

overall diameter = 60 mm

مع قطر الف لفة المحيط بالولب

$$= \text{قطر الـ ولب} + \text{ضعف الغلوص} + \text{قاسم أخذ}$$

$$= 60 + (1.5 \times 2) + 2$$

$$\approx 65 \text{ mm} \approx 2 + 3 + 60$$

بالجملة - ص

حجم الملح = 384.62 cm^3

طول الـ ولب الـ 22.8 cm - 30 cm

قطر الـ ولب 60 mm

سمانة flight thickness $4 - 5 \text{ mm}$

المسافة ~~في~~ الخلو ~~بين~~ طرف flight والعدو
أو ما يسمى Clearance

$1.5 - 2 \text{ mm}$

Shaft diameter المحسو 16 mm - 20 mm

إجهاد القص عادة $MPa 40$

المتر فاع إلى فاع على لا flight = ~~Shaft diameter~~

$$\Rightarrow \frac{2}{40} = \frac{2}{40} = \frac{20 \text{ mm}}{2}$$

سرعة خطية للفيلم عند ١٥ كيس / دقيقة
بفرض طول الحزام = 20 cm + هامش لرفع / سواك

$$= 22 \text{ cm لكل كيس}$$

سرعة خطية للحزام عند ١٥ كيس / دقيقة

$$\approx 15 \times 0.22 = 3.3 \text{ m/min}$$

~~لواختيار ضئلا اسطوانة حسب ذات قطر وفك~~

عزج الدوران المطاوب عند ٥ - ١٠ N.m

ليشمل مقاومة تدفق المطاوع والاحتكاك مع هامش أدنى

$$P = 5 \times W$$

$$\approx 5 \times 11.87 \Rightarrow P = 59.35 \text{ W}$$

هذه القيمة عند الزمن 2 ثانية
وأخترنا أقل عزوم ممكن 5 N.m

$$RPM = 60 \frac{\text{revs-per-fill}}{\text{fill times}}$$

$$RPM = 60 \frac{3.779}{2} = 113.37 \text{ RPM}$$

اللزوجة المحورية أثناء المصبة - طول الخط
الذي يتحركه المنتج بواسطة اللولب خلال المصبة

$$l_{\text{axial}} = P \propto \text{revs-per-fill} \approx 3.8 \times 0.06$$

$$\approx 0.228 \text{ m}$$

$$\approx 22.8 \text{ cm}$$

$$W = 2\pi \frac{RPM}{60} \approx 11.87 \text{ rad/s}$$

$$\Rightarrow W \approx 11.87 \text{ (rad/s)}$$

قطر العمود

$$d = \left(\frac{16T}{\pi W} \right)^{1/3} \Rightarrow d =$$

$$T = P \Rightarrow P \approx 200 \text{ W}$$

$$W = 11.87$$

$$\Rightarrow T = \frac{200}{11.87} = 16.85 \text{ N/m}$$

الواجب :

Helix angle at mean diameter

$$\phi = \arctan\left(\frac{P}{\pi D_m}\right) = 26.6^\circ$$

$$\therefore D_m = \frac{D + D_{\text{shaft}}}{2} = \frac{60 + 16}{2} = 38 \text{ mm}$$

Servo motor 200W - 300, 350W -

a. Hopper

In the hopper design, there are two parts, namely the trapezoidal prism section and the square part, as seen in Figure 2. Acrylic material was chosen as the material for making hoppers with several considerations, including strong, corrosion-resistant, and translucent. The thickness of acrylic used is 2 mm. In determining the size or dimensions and slope (angle of repose) of the hopper, you must first know the volume of fertilizer to be used.

The dimensions of the hopper include the length, width, height, and slope of the base. Dimensions: The length, width, and height are determined based on the volume of the hopper you want to make. Theoretically, the volume of the hopper can be determined by the equation:

$$V_{hp} = \frac{D_p \times A_p \times 10^3}{n_p \times \rho_p} \quad (1)$$

Information:

V_{hp}	= Volume hopper (cm ³)
D_p	= Fertilization dose (kg/ha)
A_p	= Area of fertilizing land once filling (ha)
n_p	= Number of fertilizing machine units
ρ_p	= Bulk density Fertilizer (kg/cm ³)

In determining the dimensions of the hopper, it is necessary to find the volume of the square and the volume of the decapitated prism. Because the hopper part I is square shape to find the volume using the formula:

$$V_I = p \times l \times h \quad (2)$$

Where:

V_I = Square volume

p = Panjang

l = Wide

h = Tall

Because the Hopper part II is in the form of a decapitated prism, to find Volume (V) using the formula:

$$V_{II} = \frac{1}{3} h (A_1 + A_2 + \sqrt{A_1 \times A_2}) \quad (3)$$

Where:

V_{II} = Volume of decapitated prism

A_1 = Cover Area

A_2 = Base Area

H = Tall

The basis for determining the slope (angle of repose) is to find the smallest tilt angle of the three fertilizers used, namely urea fertilizer [4].



تعديل



٢٠ / ٤

Fig. 2. Hopper design.

b. Auger Type Metering Device

In the design of the Metering Device, this is in the form of an auger, as seen in Figure 3. PVC pipe material and polyethene plastic were chosen as auger manufacturing materials with several considerations, including strength and corrosion resistance. The thickness of PVC pipes in the manufacture of augers is 1 mm and the diameter of polyethylene plastic is 1 cm.

According to [5], the theoretical volumetric capacity of an auger is formulated in Equation 1 as follows:

$$Q_t = \frac{\pi}{4} (d_{sf}^2 - d_{ss}^2) l_p n \quad (4)$$

Where:

- Q_t = theoretical volumetric capacity (m³/s)
- d_{sf} = Auger outer diameter (m)
- d_{ss} = Auger shaft diameter (m)
- l_p = pitch length (m)
- n = rotating speed Auger (rev/s)

Determination of the inner diameter and outer diameter is an essential step before determining the volumetric capacity of the auger. To measure the volumetric capacity of the auger, information about the inner diameter and outer diameter is needed using the formula: Diameter:

$$D = \sqrt{(\pi \times d_{ss})^2 + (l_p)^2} \quad (5)$$

Where:

- d = inner diameter (m)
- d_{ss} = Auger shaft diameter (m)
- l_p = Auger pitch length (m)

Outer Diameter:

$$D = D_{daun\ auger} + d - d_{ss} \quad (6)$$

Where:

- D = outer diameter (m)
- D_{daun auger} = auger leaf diameter (m)
- d = inner diameter (m)
- d_{ss} = Auger shaft diameter (m)



Fig. 3. Auger type metering device design.



3.1.2 Auger

Fertilizer rationing is a component of fertilizer equipment that functions to regulate the dropping of fertilizer in a certain amount and the desired fertilization distance. The type of metering device used is a thread-shaped auger type with material from polyline plastic and PVC pipe.

The primary determination of the dimensions of the metering device is to take the initial dimensions of a pre-existing auger-type metering device with an auger shaft diameter of 10 mm, an auger inner diameter of 11.08 mm, an auger outer diameter of 29.08 mm and a pitch length of 15 mm as the initial basis for designing dimensions as well as the size of the metering device. Then, it is mounted on the lower hopper and rotates along the metering device shaft.

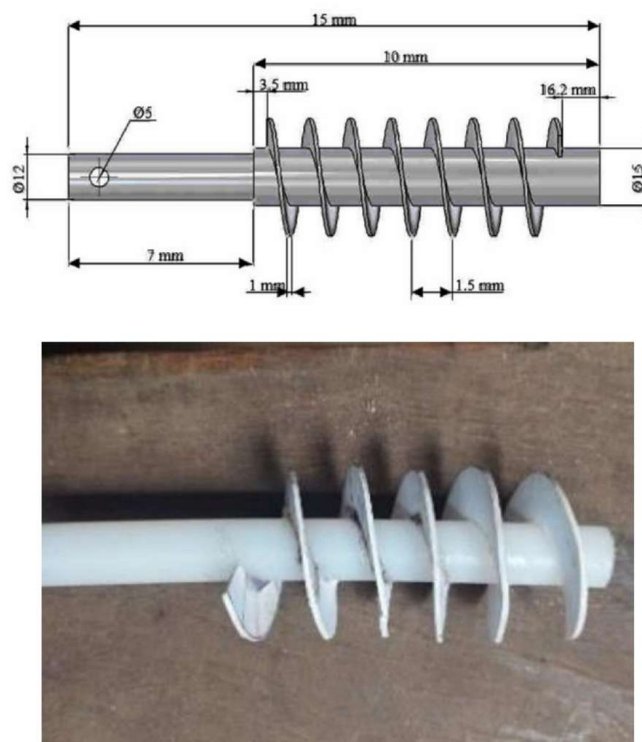
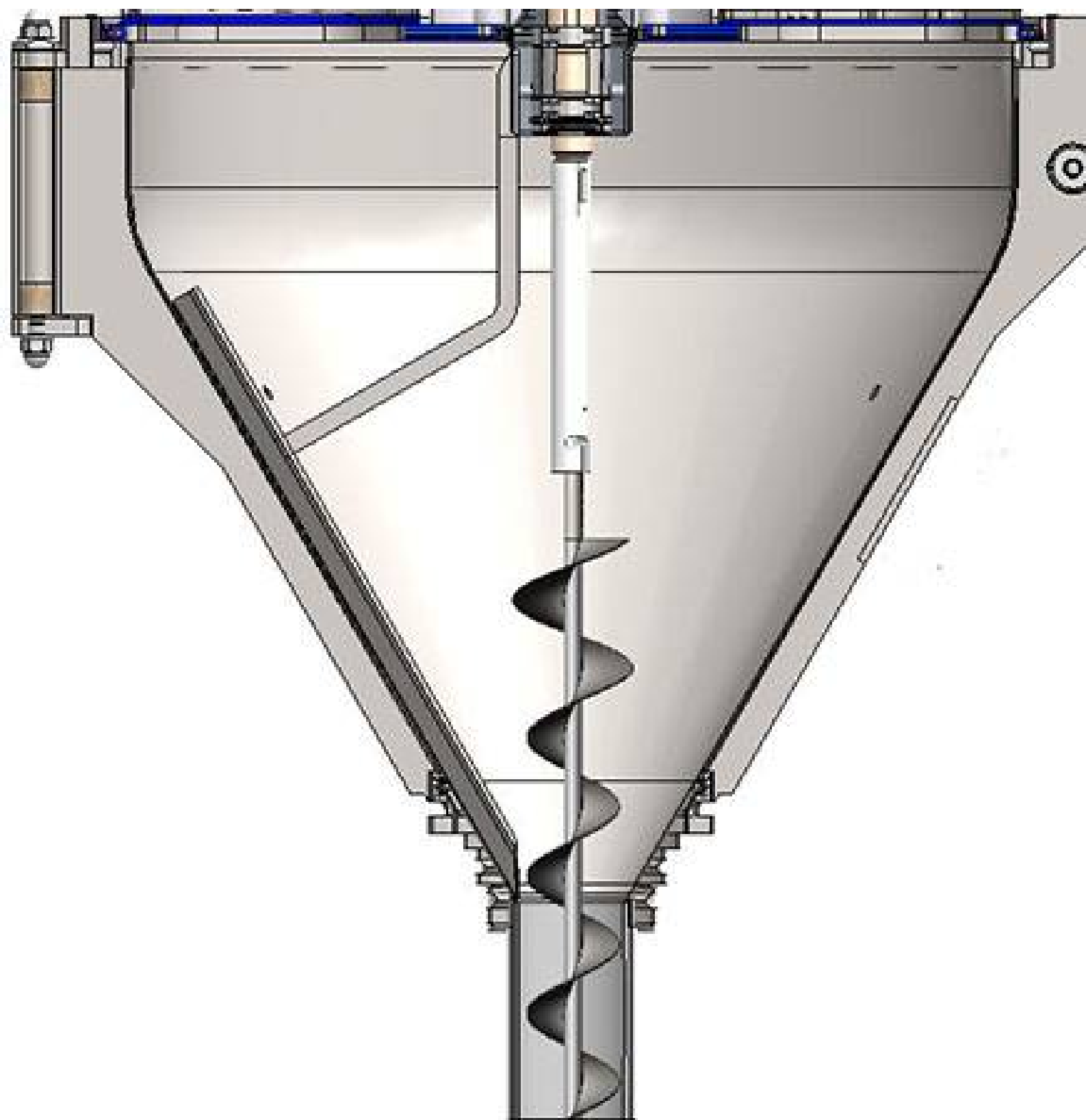


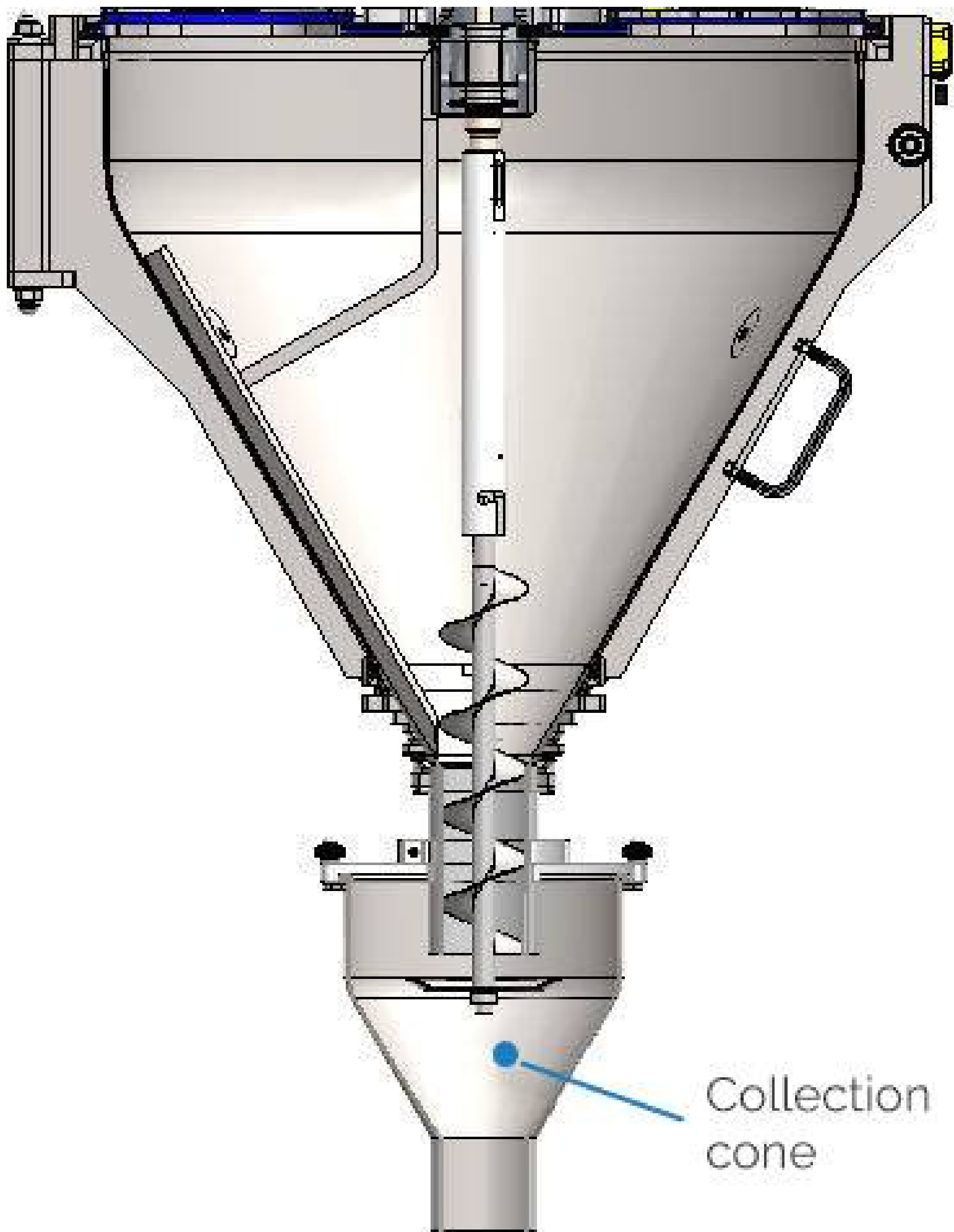
Fig. 6. Auger and auger design results on tools.

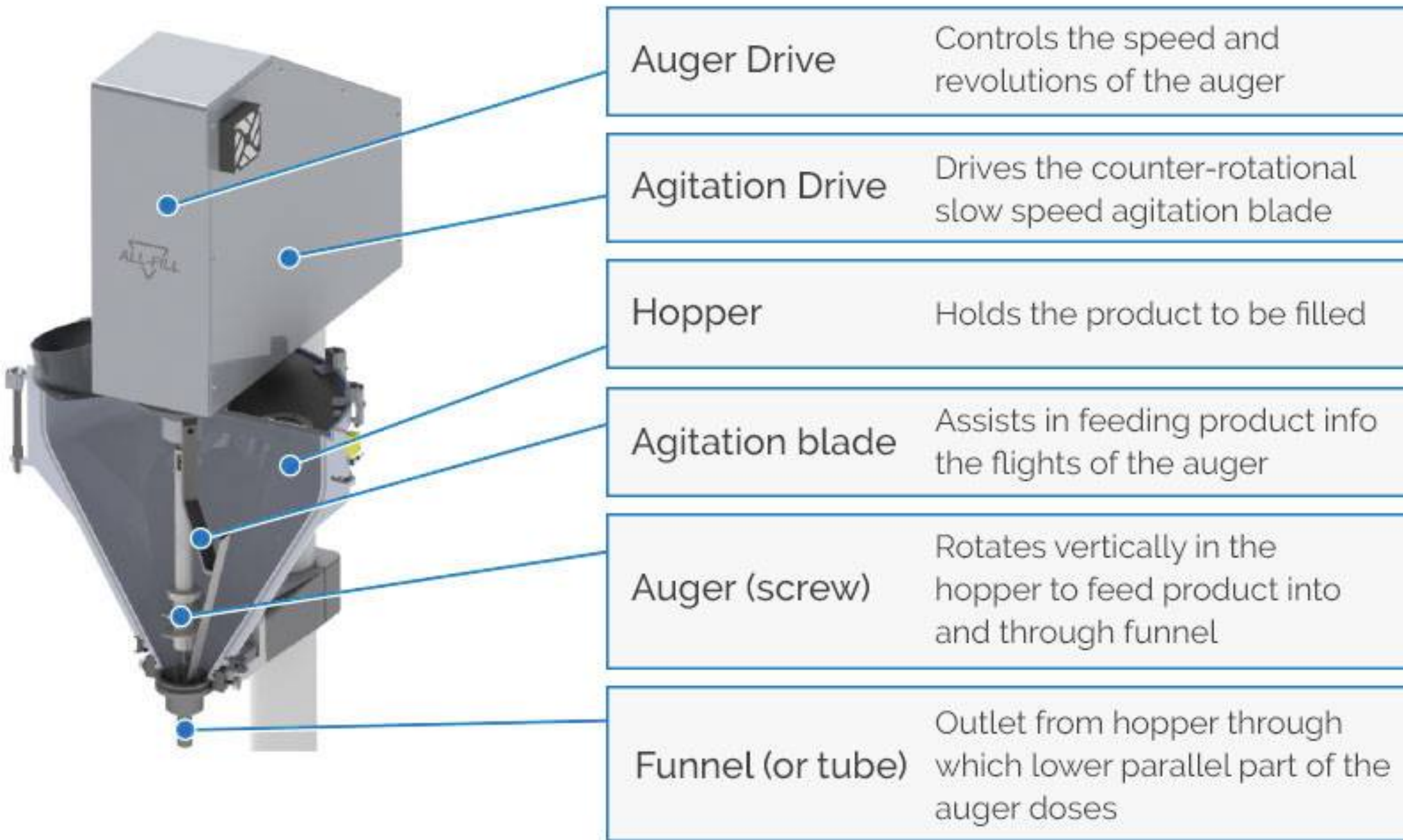
3.1.3 Control system assembly results

The design results of the DC motor speed control system can be seen in Figure 7. The hardware system in the plastic container is the control centre that sends control instructions to the DC motor. The actual speedreading results will be displayed on the serial monitor, and setting the speed set point is done by giving a value to the serial monitor. The input voltage given to the motor is a voltage of 12 Volts obtained from the power supply. The rotating motor will send speed data to the microcontroller through a rotary encoder sensor placed on the small axle shaft from the top of the motor.









motor, a stirrer, and an auger. The inner diameter of the powder container is 64 mm with a total height of 224 mm. Two air vents were added to compensate the negative pressure (vacuum) inside the container during the dosing process. In the case of a closed container, a negative pressure during powder discharge evolves, especially when dosing very fine powders where the permeability of air is low. The absence of air exchange causes a vacuum, which negatively affects the powder flowability. Three augers were used in conducting the experiments, where they have an overall length of 58 mm. Table 3.1 summarizes the geometry dimensions of the three tested augers.

Auger	Outer diameter (mm)	Inner diameter (mm)	Pitch (mm)	Auger filling volume (mm ³ /rev)	Geometry form
A	5	3.1	3	25.52	Straight
B	5.9	2.2	3.5	58.59	Straight
C	5.8	3	4.5	67.33	Conical

Table 3.1: Augers used in auger dosing experiments.

Figure 3.2 represents the geometry of the three augers with their auger filling volume per one revolution. Moreover, as an illustration example, Figure 3.2 shows the inner and outer diameters and the pitch of auger type A besides the geometries of the other two augers.

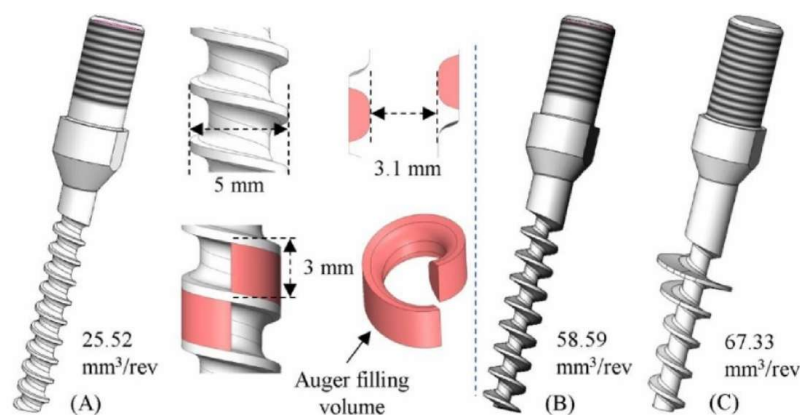
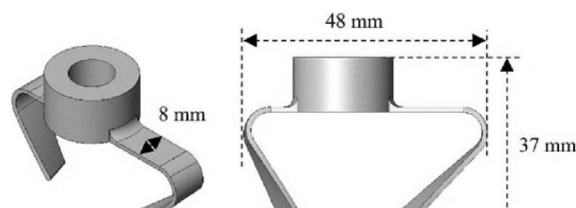


Figure 3.2: CAD models of the three augers used in the tests (A) 25.52 mm³/rev, (B) 58.59 mm³/rev, and (C) 67.33 mm³/rev.

The stirrer geometry is modeled in Figure 3.3. It has a total width of 48 mm and a total height of 37 mm. The two wings are twisted with an angle of 25°, where the inclination angle of the long wing is 45° and

that of the short wing is 57°. The advantage of using such design is the mechanism that conveys the powder downwards towards the center of the auger, and thus helping in maximizing the mass output.



product variability, a weight check may occur between bulk and trim fills. Hohmann (1988) describes the dosing mass of auger fillers as shown in Equation 1 and 2.

$$V_n = \left(\left(\frac{D_1^2 - D_2^2}{4} \right) - \left(\left(\frac{D_1 - D_2}{2} \right) \cdot b \sqrt{\left(\frac{(D_1 + D_2) \cdot \pi^2}{2} + s^2 \right)} \right) \right) \quad \text{Equation 1}$$

$$m = V_n \cdot n \cdot \rho_s \quad \text{Equation 2}$$

Where:

m = mass [lb, kg]

D_1 = outer pipe diameter [in, cm]

D_2 = auger shaft diameter [in, cm]

s = pitch [in, cm]

b = auger flight thickness [in, cm]

V_n = dosing volume [in³, cm³]

n = number of auger turns (dimensionless)

ρ_s = product density [lb/in³, kg/cm³]

In large augers, dosing errors can result from product density variability and an inability of the auger to hold product in its flights. Correct selection of an auger and accessories can improve the accuracy of the dosing volume and reduce product density variations.

Dosing errors may be reduced through the use of feedback control. This feedback control may take a form of an elementary proportional-derivative (PD) loop with a single package as a unit of time. The integral term of a proportional-integral-derivative (PID) loop is removed to avoid integral windup. Equations 3 and 4 show this simple PD loop.

$$e_i = G_{prop} * (m_i - m_s) - G_{diff} (m_i - m_{i-1}) \quad \text{Equation 3}$$

$$m_{s_{i+1}} = m_s - e_i \quad \text{Equation 4}$$

Where:

e = error term [lb, kg]

G_{prop} = proportional gain ($0 \leq G_{prop} \leq 1$, dimensionless)

m = measured mass [lb, kg]

m_s = mass set point [lb, kg]

G_{diff} = differential gain ($0 \leq G_{diff} \leq 1$, dimensionless)

Subscript i = iteration indicating current bag